MILITARY SPECIFICATION

MOUNT, HOWITZER, M178

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use within all Departments and Agencies of the Department of Defense. .

1. SCOPE

1.1 Scope. This specification covers one type of hydropneumatic gun mount for use in combat vehicles such as the M109 SPH.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards; The following Specification and Standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

BB-N-411	-Nitrogen, Technical

MILITARY

MIL-H-6083	-Hydrualic Fluid, Petroleum Base. Preservation and Operation
MIL-G-10924	-Grease, Automotive and Artillery
MIL-W-63150	-Weapons and Support Materiel Standard Quality Assurance Provisions For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn. AMSMC-QA. Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC *N/A* FSC 1025

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STANDARDS

MILITARY

MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-STD-130	-Identification Marking of U.S. Military Property

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the soliciatation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

K10895694	-Variable Recoil, Cover
K10955766	-Recuperator Cover
K12012127	-Mount Installation Howitzer, M178
K12012129	-Cradle Assembly Howitzer

PUBLICATIONS

P12012127	.	-Packaging Data Sheet for Mount, Gun, M178

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.2).

3.2 Materials. Materials shall conform to the material specifications referenced on the applicable drawings. Certification of material may be accepted in lieu of material tests. The contractor shall be responsible for determining that materials, assemblies, and parts requiring product qualifications are listed or qualified for inclusion on the applicable Qualified Products List (QPL). In the instance that a qualified vendor no

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longer exists on the QPL, the contractor is responsible for selecting a vendor whose product meets the mininmum requirement of the specification. The contractor must also submit to the government for approval documentation which confirms the vendors ability to meet the specification in question Contractor's inspection record shall specifically list such components, name of supplier(s), and number and date of applicaable QPL or governement approved vendors not on a QPL.

3.3 Construction. The mount, and all of its assemblies and Components, shall be constructed and assembled as specified on the applicable drawing referenced on DrDrawing K12012127. Dimensions and tolerances shall be as specified on the drawings

3.3.1 Hydraulic system

3.3.1.1 Particle size. The recoil mechanism shall be filled with hydraulic fluid, bled and refilled until all air and measurable moisture are eliminated. The initial and subsequent filling of the hydraulic system shall be with hydraulic oil conforming to MIL-H-6083 and filtered through a 5 micron filter. Within five minutes after mechanical gymnastication, a minimum of one pint of oil shall be removed from the lowest drain hole from each recoil cylinder. Sample shall be drained into a suitable container. Particles in each sample shall not exceed the limits specified in Table I.

TABLE I. Particle limits.

Particles in either sample shall not exceed the following:

Metallic

Ferrous - 40 microns and a maximum of two partitles over 40 microns but not exceeding 200 microns in greatest dimension?

Non-Ferrous - 200 microns and a maximum of two particles over 200 microns but not exceeding 400 microns in greatest dimension.

Non-Metallic

Abrasive - 200 microns and a maximum of two particles over 200 microns but not exceeding 400 microns in greatest dimension.

Non-abrasive - No limit, however, close surveillance to be maintained to insure minimum quantity.

Lint/Fibers - 600 microns x 60 microns and a maximum of eight particles over 600 microns x 60 microns but not exceeding 3000 microns x 200 microns.

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3.3.1.2 Leakage. During cycling and after all cycling, no indication of unusual hydraulic leakage shall occur.

3.3.2 Surface cleanliness. All finish machined surfaces shall be free of condensation or other foreign residues from the time of final machining until final assembly; All parts and assemblies being assembled into the completed mount shall be free of all foreign material after assembly. Assembly shall be performed in the controlled environment specified in 6.3.

3.3.3 Part modification. No parts shall be altered or handfitted during assembly unless such modification has been authorized by an approved-drawing change, or is authorized by instructions presently listed on the drawings.

3.3.4 Air, gas and accumulator system. The gas system shall be clean and free of all foreign matter. The hydroscopic content. of the air used to blow the lines shall not exceed one percent of the total volume of the air used. The system shall be filled, bled, and refilled until all air and measurable moisture are eliminated.

3.3.5 Cylinder pressure. The cylinders of the mount shall withstand the pressures specified on the drawings without damage, permanent deformation or leakage.

3.3.6 Pistons, rods, and nuts or collars. The pistons, piston rods, and nuts or collar shall withstand the pull test specified on the drawings without permanent set or failure of the screw threads.

3.3.7 Lubrication. Lubricants corformint to MIL-G-10924 shall be used for the assembled mount.

3.3.8 Gas. The nitrogen used to charge and fill the nitrogen chamber shall conform to BB-N-411. The charging pressure shall be specified on Drawing 12012127. .

3.4 Performance.

3.4.1 Mechanical gymnastication. Each mount shall be subjected to the mechanical gymnastication test as specified in 4.6.1 without evidence of erratic movement and shall meet the requirements of 3.4.1.1 and 3.4.1.2. During and after gymnastication, the gun mount shall meet the requirement of 3.4.5 and 3.4.6.

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3.4.1.1 Counter recoil (CTRR) time. The CTRR for each mechanical gymnastication shall not exceed the requirements specified below.

Elev. in Degree	CTRR Stroke Time (Seconds)	Buffer Stroke Time (Seconds)
0°	3.500 - 5.500	3.000 - 5.000
45° Long Recoil	3.500 - 8.500	3.000 - 8.000
45° Short Recoil	3.500 - 6.500	3.000 - 6.000
Max imum	3.500 - 7.000	3.000 - 6.500

The gymnasticator is set for 15 ± 1 second pull back time at 0° elevation and 36 inches. This setting is not changed for other elevation simulations.

3.4.1.2 operating rods. All operatomg rods shall be inspected for scratches during and after mechanical gymnastication is completed. Continued evidence of rod scratching will result in teardown of the affected cylinder, Polishing out the scratch or replacing of the rod, replacing of the seal and cleaning and inspection of the cylinder assembly components. Light scratches that can be polished out and that do not re-occur with continued cycling are not cause for teardown.

functional firing test as specified in 4.6.2 without evidence of malfunction or failure and shall meet the criteria of 3.4.2.1 through 3.4.2.3. During and after functional firing, the gun mount shall meet the requirement of 3.4.5, 3.4.6 and 3.4.7.

3.4.2.1 Counter-recoil (CTRR].

3.4.2.1.1 CTRR velocity. At 1 inch from the complete in-battery position, the CTRR velocity shall not exceed one and one-half (1 1/2) feet per second.

3.4.2.1.2 CTRR cycle time. The CTRR cycle time in seconds shall be:

Simulated Elevation	Above 30° F	30° F and Below
0°	.700 - 1.9.00	.700 - 3.500
Maximum	.900 - 3.900	.900 - 5.900

3.4.2.2 Recoil cycle. The recoil cycle travel (in inches) and total cycle time (in seconds) shall be as follows:

3.4.2.2.1 Recoil travel in inches.

Simulated Elevation	Recoil Travel
0°	27 - 36
Max imum	18 - 27

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3.4.2.2.2 Total cycle time in seconds.

Simulated Elevation	Above 30° F	300 F and Below
0°	.900 - 2.000	.900 - 3.600
Maximum	1.000 - 4.000	1.000 - 6.000

3.4.2.3 Recoil performance zones. For all 100% acceptance test cvcles, actual recoil cycling oil pressure versus time curves shall be within defined recoil performance zones shown in Figures 1 and 2.

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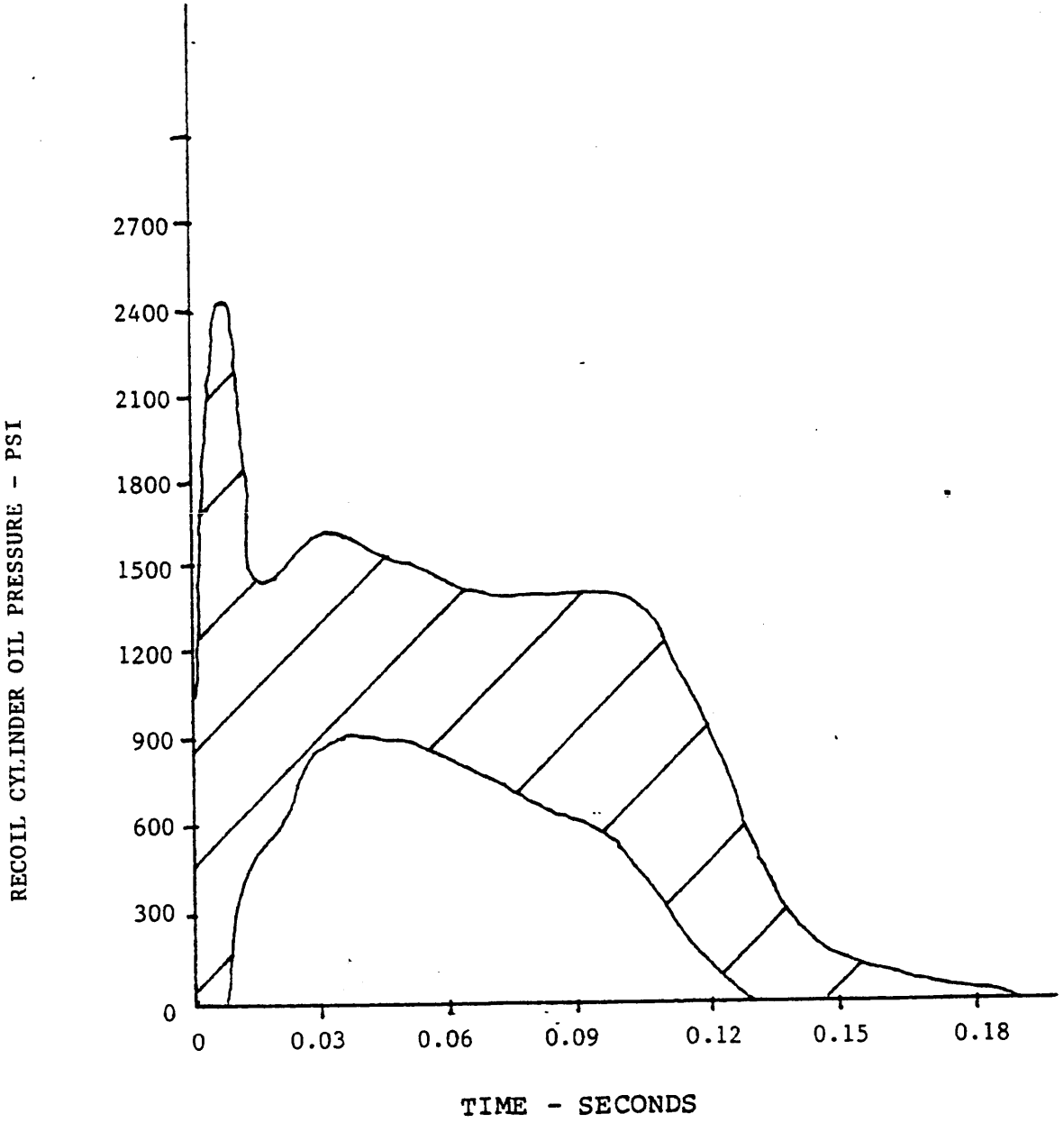


FIGURE 1. FUNCTIONAL FIRING

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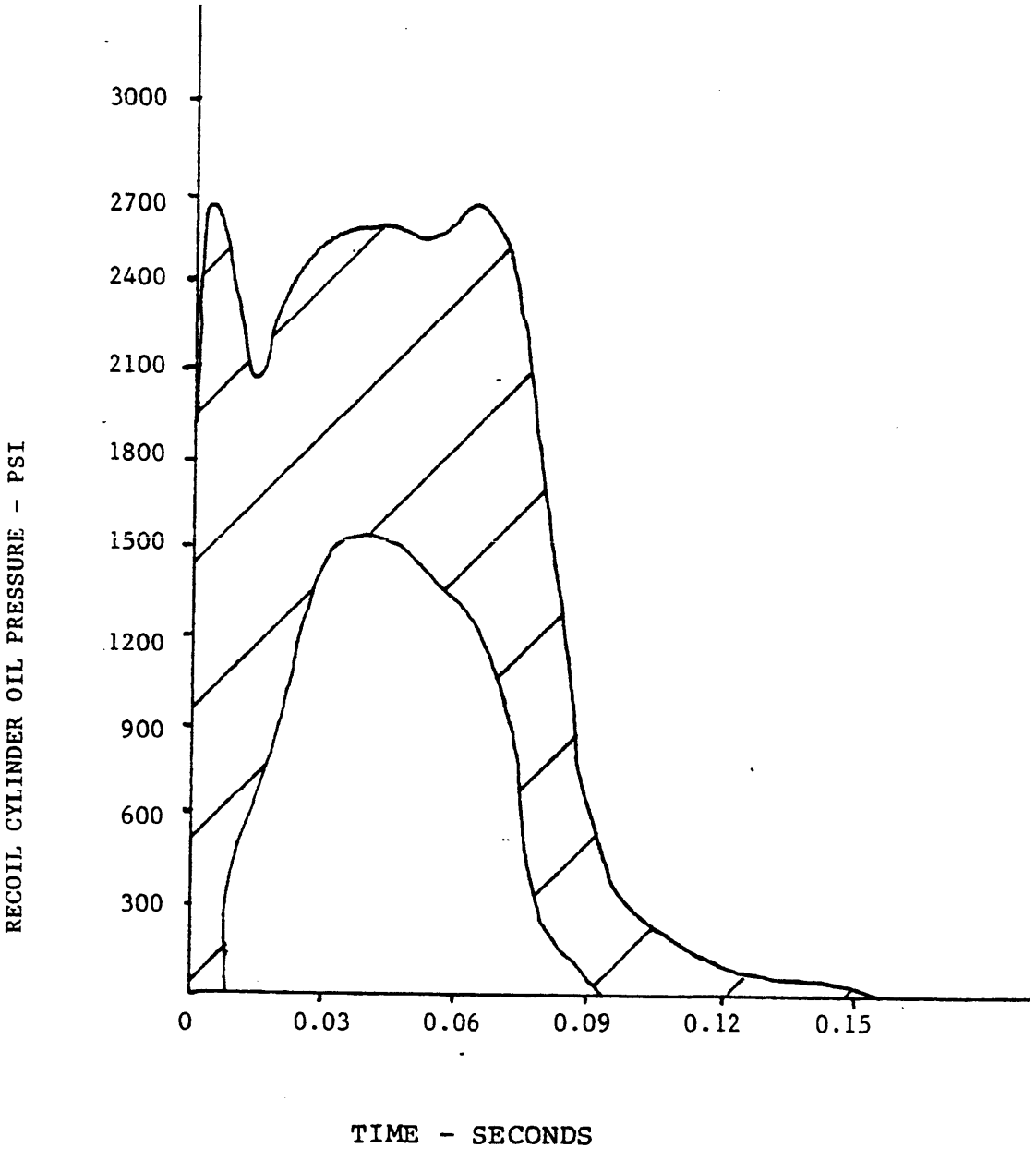


FIGURE 2. SHORT RECOIL - FUNCTIONAL FIRING

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3.4.3 Proof firing. Each sample mount shall be subjected to proof firing at a Government proving ground as specified in 4.6.3 without evidence of malfunction or failure and shall meet the criteria of 3.4.3.1 through 3.4.3.3. During and after proof firing, the gun mount shall meet the requirements of 3.4.5, 3.4.6 and 3.4.7.

3.4.3.1 CTRR velocity. At one inch from the complete in-battery position, the CTRR velicuty shall not exceed one and one-half (1 1/2) feet per second.

3.4.3.1.2 CTRR cycle time. The CTRR cycle time at 400F to 100° F in seconds shall be:

Elevation in Degrees	CTRR Stroke
0°	.650 - 1.250
Maximum	.850 - 1.850

3.4.3.2 Recoil cycle. The recoil cycle travel (in inches) and total cycle time (In seconds) shall be as follows:

3.4.3.2.1 Recoil travel in inches.

Elevation	Recoil Travel
0°	27 - 36
Max imum	18 - 27

3.4.3.2.2 Total cycle time in seconds.

Elevation	Total Cycle Time
0°	.800 - 1.400
Maximum	1.000 - 2.000

NOTE : Conditions of having-out-of-battery at the higher elevations are acceptable on the M178 Gun Mount provided it returns to battery upon lowering tube to loading elevations (approximately 110 mils). The test agency is to specify length of hanging-out-of-battery's in reports.

3.4.3.3 Recoil performance zones. For all full charge firings actual recoil cyclinder oil pressure versus time curves shall be within defined recoil performance zones shown in Figures 3 and 4.

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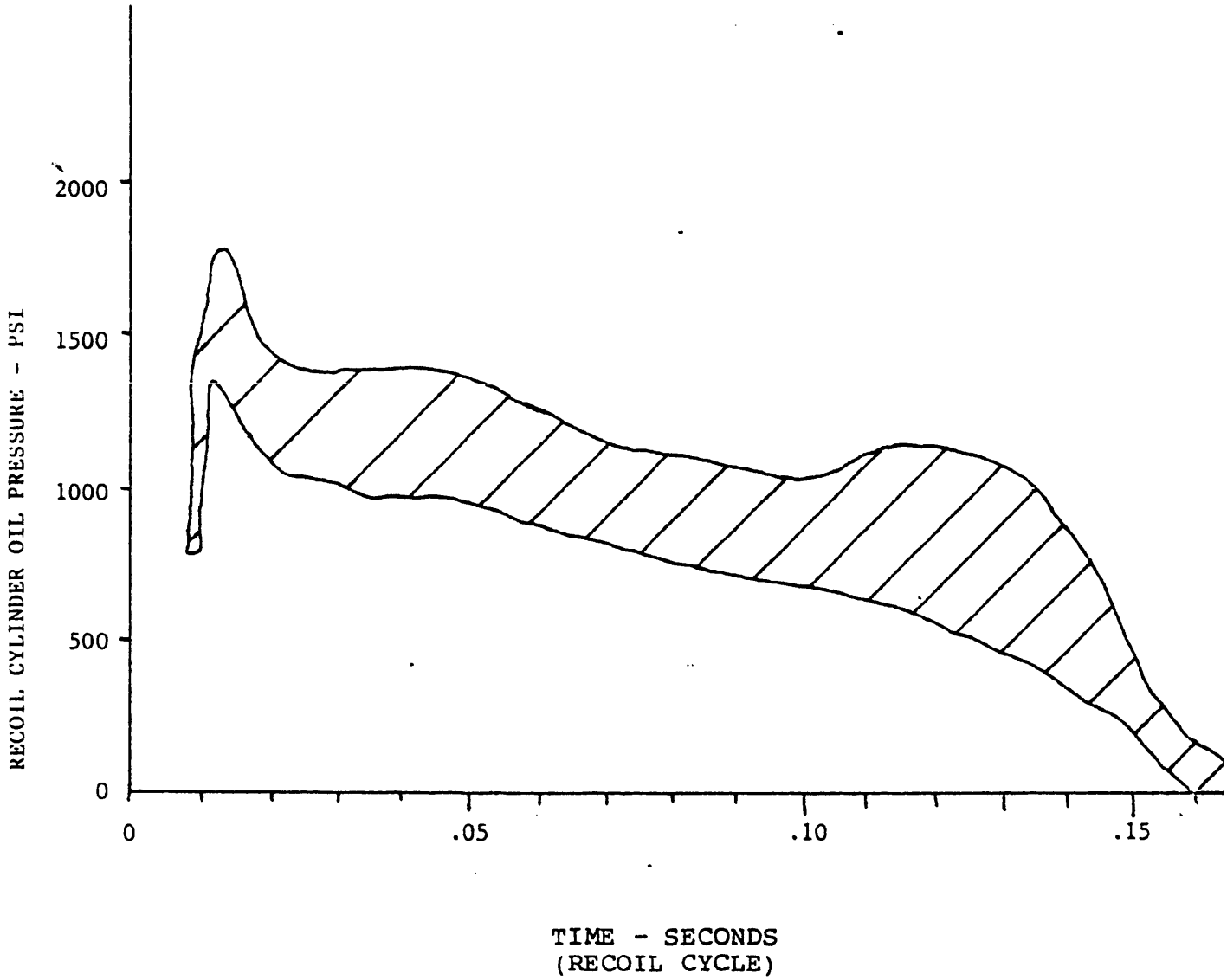


FIGURE - PROOF FIRING

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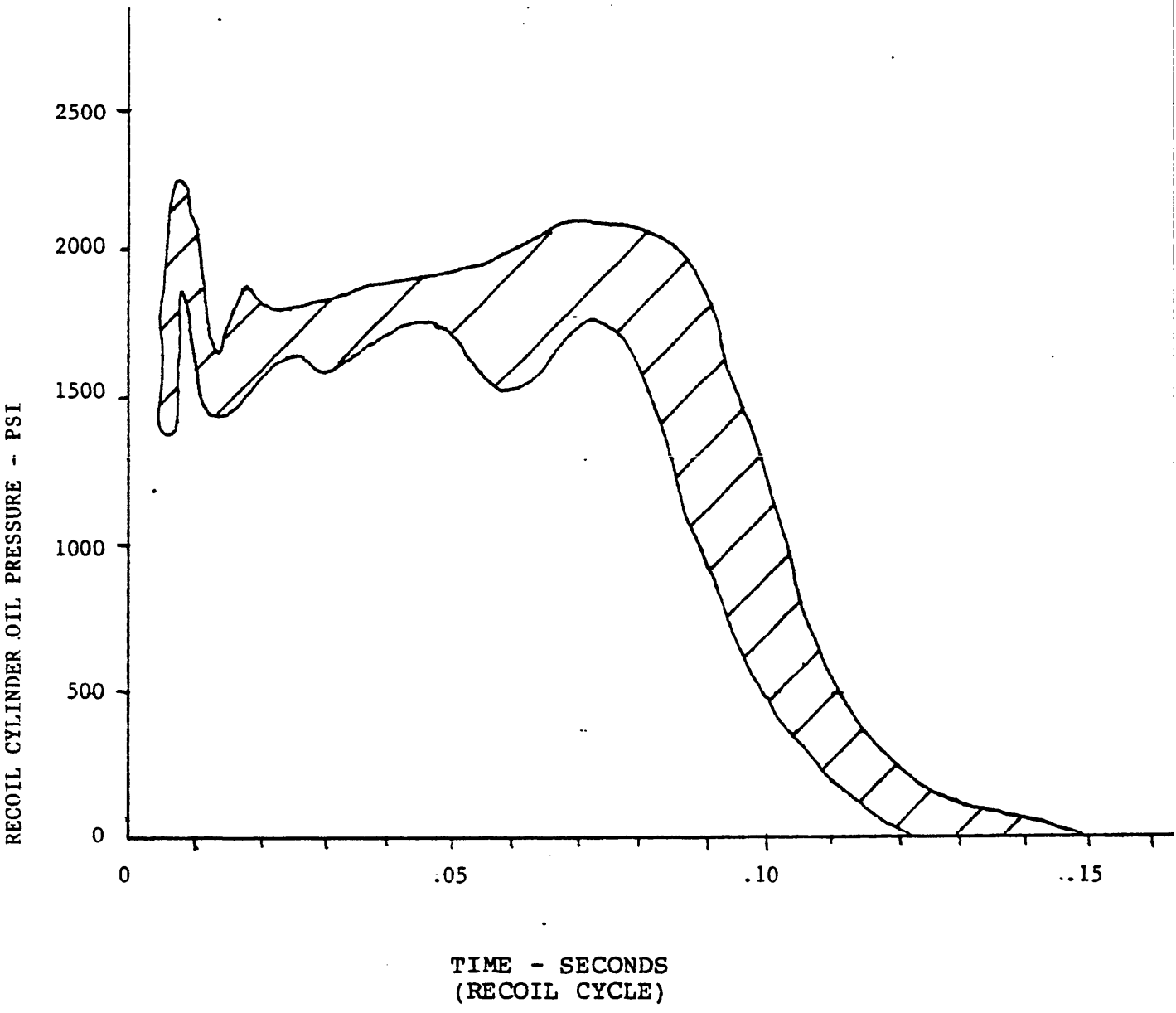


FIGURE 4 - PROOF FIRING

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3.4.4 Climatic conditions. A sample as specified in 4.5.1 shall be proof fired as specified in 3.4.3 at each of the following temperatures: +125°F and -50°F.

3.4.5 Hydraulic leakage.

3.4.5.1 Dynamic. Immediately following cycling of the gun mount there shall be no more than one milliliter (ml) of oil accumulation per every five (5) cycles tested at any seal interface location. Immediately following this determination all oil accumulations are to be wiped clean prior to an additional five (5) cycle incremented test.

3.4.5.2 Static. After the gun mount has set statically for 24 hours after completion of each test, (mechanical gymnastication, functional firing, proof firing) there shall be no more than 1 ml of oil accumulation at any seal interface location and no evidence of seepage of hydraulic fluid from any of the hydraulic tubes, manifolds, hose connections, and fill/bleeder valves.

3.4.6 Nitrogen leakage. Mounts shall also be examined for leakage of nitrogen. Any indications of foamy or emulsified condition of reserve oil indicates gas leakage and shall be cause for rejection.

3.4.7 Correlation of data. The data recorded for 3.4.2, functional firing, and for 3.4.3, proof firing, shall be correlated. If the correlation between length of recoil and cycle time is not satisfactory, as determined by the procuring agency, the mounts shall be returned to the contractor.

3.5 Protective finishes.

3.5.1 Inorganic. Inorganic finishes, including metal deposited coatings and surface treatments, shall be as Specified on the drawings.

3.5.2 Painting. Painting shall be as specified on the drawings.

3.6 Marking. Identification marking, including application of part numbers, and instruction marking shall be as specified on the drawings and in MIL-STD-130.

3.7 Data requirements. The contractor shall generate data in accordance with the requirements of the data item description cited in 4.6.5.

3.8 Workmanship. Each mount shall be free of defects that affect overall appearance, serviceability, or performance such as burrs, fins, sharp edges, and surface flaws or blemishes.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform' to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet the requirements of Section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions as used In this specification are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.4 First aritcle.

4.4.1 First article sample. A first article sample shall be selected and submitted for testing at a location approved by the Government in accordance with the contract requirements (see 6.2). The sample shall be representative of the production processes to be used during quantity production.

4.4.2 First article inspection. The first article sample inspection shall be Inspected to determine conformance with the requirements of the drawings and this specification.

4.4.3 First article rejection. If any sample fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of the sample to comply with the requirements of this specification.

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4.5 Quality conformance inspection.

4.5.1 Sampling. Each mount shall be inspected for compliance to this specification except sampling for proof firing as specified in 4.7.3 shall be the first three production mounts, and one from each two month's production or one from each twenty-five produced. In addition, the first three production mounts shall be climatically fired as specified in 3.4.4. Failure to comply with the specified requirements shall be cause for rejection.

4.5.2 Examination.

4.5.2.1 Materials. Materials used in the manufacture of mounts shall be examined and tested in accordance with the material specifications referenced on the applicable drawings. The contractor's inspection records shall be checked to determine conformance to 3.2.

4.5.2.2 Hydraulic system. To determine compliance with 3.3.1, a minimum of one pint of oil shall be removed from each recoil cylinder of each mount. The sample shall be taken within five minutes after gymnastication. Evidence of particles exceeding 3.3.1 shall be cause for rejection. Additionally, failure to comply with the hydraulic leakage requirements of 3.3.1 and 3.4.5 shall be cause for rejection.

4.5.2.3 Surface cleanliness. Each mount shall be examined to determine compliance with the cleanliness requirements specified in 3.3.2. Failure to comply with 3.3.2 shall be cause for rejection of the mount.

4.5.2.4 Part modification. Each mount shall be examined to determine compliance with requirements specified in 3.3.3. Any evidence of parts being altered or hand-fitted during assembly beyond the provisions of 3.3.3, shall be cause for rejection of the mount.

4.5.2.5 Gas and accumulation system. To determine conformance to 3.3.4, the system shall be examined for cleanliness, air and moisture. Failure to comply with 3.3.4 shall be cause for rejection of the mount.

4.5.2.6 Cylinder pressure. To determine compliance with 3.3.5, the cylinders of each mount shall be pressurized in accordance with drawing requirements and inspected for damage, permanent deformation or leakage. Pressures shall be measured with Government-approved gages. Failure to comply with 3.3.5 shall be cause for rejection of the mount.

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4.5.2.7 "Pistons, rods and nuts or collars. To determine compliance with 3.3.6, the pistons, piston rods, and nuts or collars of each mount shall be inspected for permanent set or damaged threads after being pull-stressed by Government approved equipment. Failure to comply with 3.3.6 shall be cause for rejection. of the mount.

4.5.2.8 Lubrication. Inspection and tests for lubricants shall be in accordance with MIL-G-10924. Failure to comply with MIL-G-10924 shall be cause for rejection of the mount.

4.5.2.9 Gas. Inspection and tests for nitrogen shall be in accordance with BB-N-411. Failure to comply with BB-N-411 shall be cause for rejection of the mount.

4.5.2.10 Visual examination. Each mount shall be examined for the following defects. One or more defects shall be cause for rejection of the mount.

a. Incorrect assembly or missing parts

b. Discontinuity or absence of protective finish

c. Incorrect or missing marking

d. Color of paint not as specified

e. Workmanship not as specified.

4.5.2.11 Packaging and packing. Examination of packaging and packing shall be in accordance with Packaging Data Sheet 12012127.

4.6 Testing.

4.6.1 Mechanical gymastication. Each mount shall have a mechanical gymnastication in accordance with test method 4.7.1 to determine compliance with requirements specified in 3.4.1.

4.6.2 Functional firing. Each mount shall be fired in accordance with test method 4.7.2 to determine compliance with requirements specified in 3.4.2.

4.6.3 Proof firing. Mounts shall be proof fired as speified in 4.7.3 to determine compliance -with requirements specified in 3.4.3. In the event a Gun Mount sample fails to satisfy acceptance criteria, the following procedure will be implemented:

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a. 'The failed sample, accompained with a detailed report as to the deficiencies found, will returned to the contractor for immediate corrective action to the represented lot and subsequent production. No dismantling or-altering of the mount will occur at the Government test facility without the documented consent of the contractor.

b. The corrected lot shall not be released until such time as three additional mounts selected by the Government consecutively satisfy acceptance criteria (3.4.3) when tested at the Government proving ground.

Three samples from the following production lot will be submitted to the Government proving ground for acceptance testing.

(1) Acceptance of these three (3) mounts shall permit return to the normal sampling procedure of one mount from each two month's production or one mount from each 25 mounts produced, whichever occurs first.

(2) Rejection of any one of the three (3) mounts tested shall require repeating of steps a, b and c.

4.6.5 Data requirements. The contractor shall submit a quality inspection test, Demonstration and Evaluation report in accordance with Data Item Description DI-R-1724 (see 6.2.2) giving the detailed test results for 4.6.1, 4.6.2, and 4.6.3.

4.7 Methods of inspection.

4.7.1 Mechanical gymastication. Each mount, including the cannon or simulated equal shall be gymnasticated on a Government-approved gymnasticator for a minimum of 40 cycles as follows:

a. 10 cycles at O°, long recoil (36 inches)

b. 10 cycles at 45°, long recoil (36 inches)

c. 10 cycles at 45°, short recoil (24 inches)

d. 10 cycles at max. QE, short recoil (24 inches)

Smooth counter-recoil performance is required prior to completion of gymnastication. Recuperator nitrogen charge pressure shall be set at 700 ± 10 Psi at 700 F temperature prior to gymnastication.

4.7.1.1 Particle count. Ater gymastication, oil shall be drained from the assembly for particle count (3.3.1).

4.7.1.2 Hydraulic leakage. During each During each 10 cycle test, the mount shall be inspected for evidence of hydraulic leakage (3.4.5).

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4.7.1.3 -Nitrogen leakage. During each 10 cycle test, the mount shall be inspected for evidence of nitrogen leakage (3.4.6).

4.7.2 Functional firing. Each gun mount shall be submitted to the powder gymnastication or hydraulic artillery simulation test as specified in Table II with the recuperator nitrogen pressure "set to 700 ± 25 psi at ambient temperature prior to firing. All testing shall be done on a test stand at zero elevation with the variable recoil locked for long recoil or short recoil as applicable to simulate specified elevations. The gymnaSticator/simulator shall be varied as required to meet the stated impulse levels. All measurements required by 3.4.2 shall be made for erratic motion in counter-recoil. During and after firing, each mount shall be examined to determine compliance with oil leakage requirements specified in 3.4.5 and nitrogen leakage as specified in 3.4.6.

TABLE II. Functional firing

Round Number	Simulated Elevation	Recoil Mode	Impulse Level
1	0°	Long	50%
2	0°	Long	100%
3	Max.	Short	100%
4	Max.	Short	100%
5	0°	Long	100%

NOTE : Implulse applied to the recoil mechanism at 100% implulse. level is 6350 pound-second mimimum for long recoil and 6800 pound-second mimimm for short recoil. At least one cycle in both long and short mode shall meet or exceed the 100% level. The recoil oil pressure for long recoil and for short recoil should be within the performance zones specified by figures 1 and 2 at the 100% level.

4.7.3 Proof firing. Mounts selected as specified in 3.4.3 and 3.4.4 shall be proof fired as specified in Table III. Recuperator nitrogen pressure shall be set at 700 ± 25 psi at ambient temperature prior to firing. All measurements required by 3.4.3 shall be recorded for each firing. Observation shall be made for erratic motion in counter-recoil. During and after firing each mount shall be examined to determine compliance with oil leakage requirements specified in 3.4.5 and nitrogen leakage as specified in 3.4.6.

TABLE III. Proof firing

Round Number	Elevation	Recoil Mode	Charge
1	0°	Long	M4A2, Zone 6
2	0°	Long	M119, Zone 8
3	Max.	Short	M119, Zone 8
4	Max.	Short	M119, Zone 8
5	00	Long	M119, Zone 8

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NOTE : All M119 charges shalll be reconditioned for 24 hours to 145° F prior to proof firing and a projectile weighing 104± 1/2 lb. shall be used.

4.8 Inspection equipment. See MIL-W-63150.

5. PACKAGING

5.1 Pilot pack. A pilot pack consisting of a complete and packed shipping container, packaged in accordance with Packaging Data Sheet P12012127 to the level of protection specified (see 6.2), shall be forwarded as specied in the contract.

5.2 Preservation, packaging, packing and marking. The mounts shall be unit packaged, packed and marked In accordance with the requirements of Packaging Data Sheet P12012127 to the level of protection specified (see 6.2).

5.3 Repair parts. Repair parts shall be unit packaged, packed and marked in accordance with the requirements of the applicable Packaging Data Sheets (see 6.2) .

6. NOTES

6.1 Intended use. This type of hydropneumatic gun mount is designed for use in combat vehicles such as the M109 SPH.

6.2 Ordering data.

6.2.1 Acquisition requirements.

a. Title, number and date of this specification.

b. Title, number and date of applicable documents and drawings.

Requirement and the number of mounts to be submitted for first article by the contractor (see 3.1 and 4.4).

d. Place of inspection and test.

e. Levels of packaging required (see S). ,

f. Air conditioning for assembly (see 6.3).

6.2.2 Data requirements. When this specification is used in an acquisition which Incorporates a DD Form 1423, Contract Data Requirement List (CDRL) , the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD Far Supplement Part 27, Sub-part 24.410-6 (DD Form 1423 are invoked.

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used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Date Requirements	Applicable DID
4.6.1, 4.6.2, 4.6.3	Quality Inspection Test, Demonstration and Evaluation Rpt.	DI-R-1724

6.3 Assembly areas. Assembly areas, special provisions, and equipment should be provided by the contractor. -The assembly areas should be air conditioned using an air conditioning unit with a 50 micron filter. The area temperature should be maintained between 680 and 72°F. The air flow rate per cubic inch per minute should be such as to maintain area air pressure at between .75 and 1.5 inches of water. Relative humidity should not exceed 40-50 percent. Areas should be clean and free of dust, lint, and abrasive materials that may become entrapped in the assembled mechanisms. No grinding, - filling, scraping, or any operation producing abrasive dust should be performed in the assembly area.

6.4 Functional firing. Information regarding the artillery simulator, see 4.7.2, is available from the Commanding Officer, Rock Island Arsenal, Rock Island, IL 61299-6000-.

6.5 Hydraulic leakage. The quantity of gun mount bydraulic fluid leakage cannot be precisely measured, therefore the judgement and experience of the inspector will be relied upon for compliance with the criteria of 3.4.5. One millilter of hydraulic fluid is composed of approximately 25 drops.

6.5.1 Seal interface locations. The seal interface locations are defined as the area around the following components:

a. Buffer Rod Wiper.

b. Recoil Rod Wiper, (2 places).

c. Recuperator Rod Wiper.

d. Recuperator Head and. Guide Pins (2 places).

e. Variable Recoil Selector Gears, under the Variable Recoil Cover, P/N 10895694, (2 places).

f. Recuperator Piston and Guide Pins (2 places), under the Recuperator Cover P/N 10955766.

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6.5.2 Hydraulic fluid seepage. Hydrualic fluid see page is defined as any evidence of hydraulic fluid beyond the seal that does not result in the formation of a droplet.

6.6 Subject term (key word) listing.

"Combat vehicle
Functional firing
Gun mount
Gymnastication
Howitzer
Hydraulic system
Hydropneumatic
Proof firing
Recoil

Custodian:
Army AR

Preparing activity
Army - AR

(Project 1025-A134)

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MIL-M-63988	MOUNT, HOWITZER, M178

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5. PROBLEM AREAS

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